STATISTICS & MACHINE LEARNING · LESSON 26 OF 28

Neural Networks and Deep Learning: From a Neuron to Transformers

Understand the models behind modern AI. Neurons, weights and activation functions, forward pass and backpropagation, a two-layer network written from scratch in NumPy, multilayer perceptrons in scikit-learn on images and tabular data, training practice (learning rate, batches, epochs, early stopping, dropout), and a map of CNNs, RNNs, transformers and LLMs with PyTorch and Keras code.

ML track (added) · GenAI bridge | ✦ scikit-learn 1.7+ · PyTorch 2.x · Keras 3 | Advanced | ⏱ 130 min

By the end of this lesson you can

- Explain a neuron, layers, activation functions and the forward pass
- Explain loss, gradients and backpropagation, and train a small network from scratch
- Train and tune MLPs with scikit-learn and compare them with classical models
- Apply practical training techniques: scaling, learning rate, batching, early stopping, regularisation
- Describe CNNs, RNNs and transformers and where each is used

Datasets used

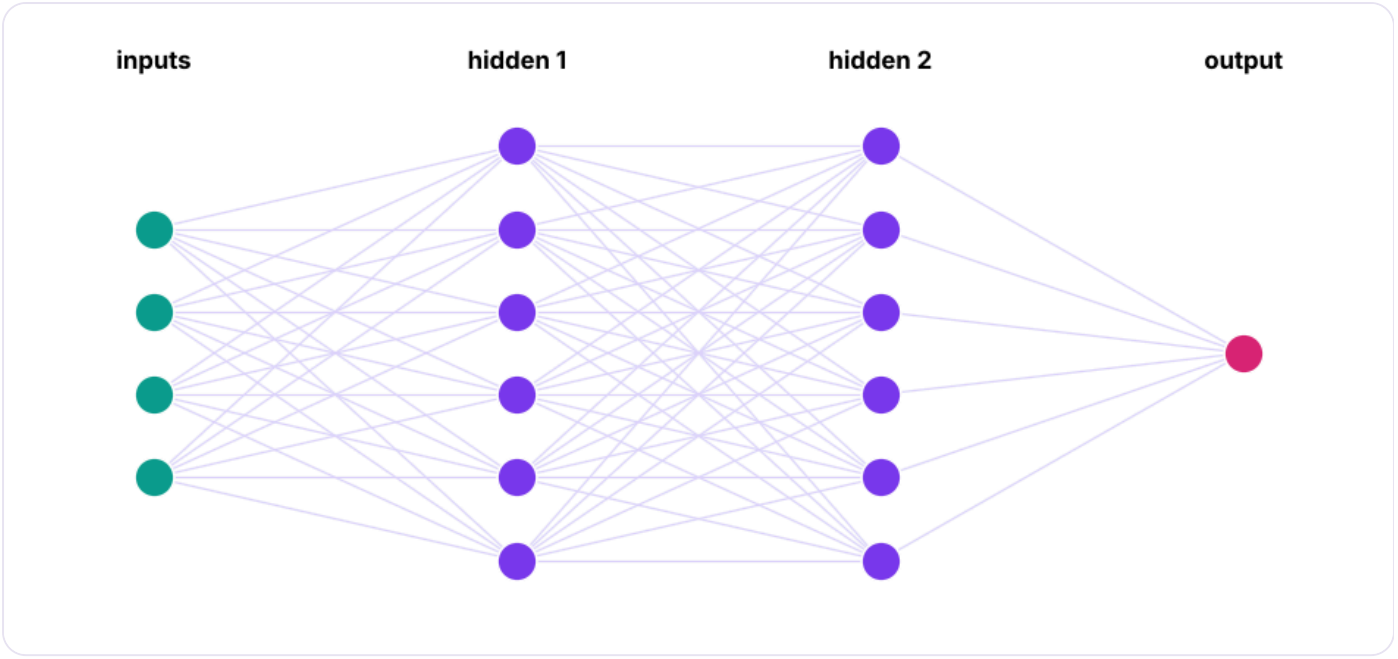
`telecom_churn.csv`

A neuron

An artificial **neuron** computes a weighted sum of its inputs plus a bias, then applies an **activation function**:

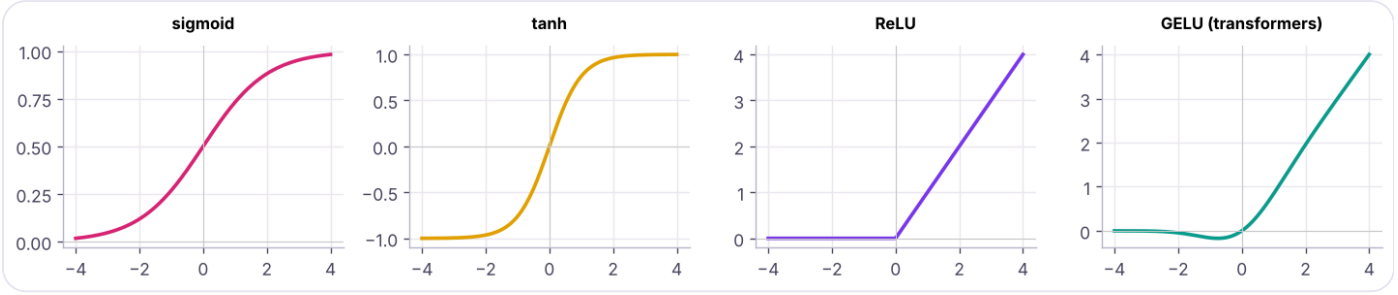
$$a = f\left(\sum_j w_j x_j + b\right)$$

With the sigmoid activation, one neuron **is** logistic regression. Stacking neurons in **layers** lets the network learn non-linear patterns.



A multilayer perceptron: inputs, two hidden layers and an output. Every connection has a weight learned from data.

Activation functions



Common activation functions. ReLU is the default for hidden layers; sigmoid and softmax are used for outputs.

Activation	Where	Why
ReLU (max(0, z))	Hidden layers	Simple, fast, avoids vanishing gradients
GELU, SiLU	Hidden layers of transformers	Smooth variants of ReLU
Sigmoid	Binary output	Probability between 0 and 1
Softmax	Multiclass output	Probabilities that sum to 1
None (linear)	Regression output	Any real value

Without non-linear activations, any stack of layers collapses into a single linear model.

How networks learn: backpropagation

1. **Forward pass:** compute predictions layer by layer.
2. **Loss:** measure the error (cross-entropy for classification, MSE for regression).
3. **Backward pass:** use the chain rule to compute the gradient of the loss with respect to every weight (**backpropagation**).
4. **Update:** gradient descent step on every weight (lesson 11), usually with the **Adam** optimiser on mini-batches.
5. Repeat for many **epochs** (passes over the data).

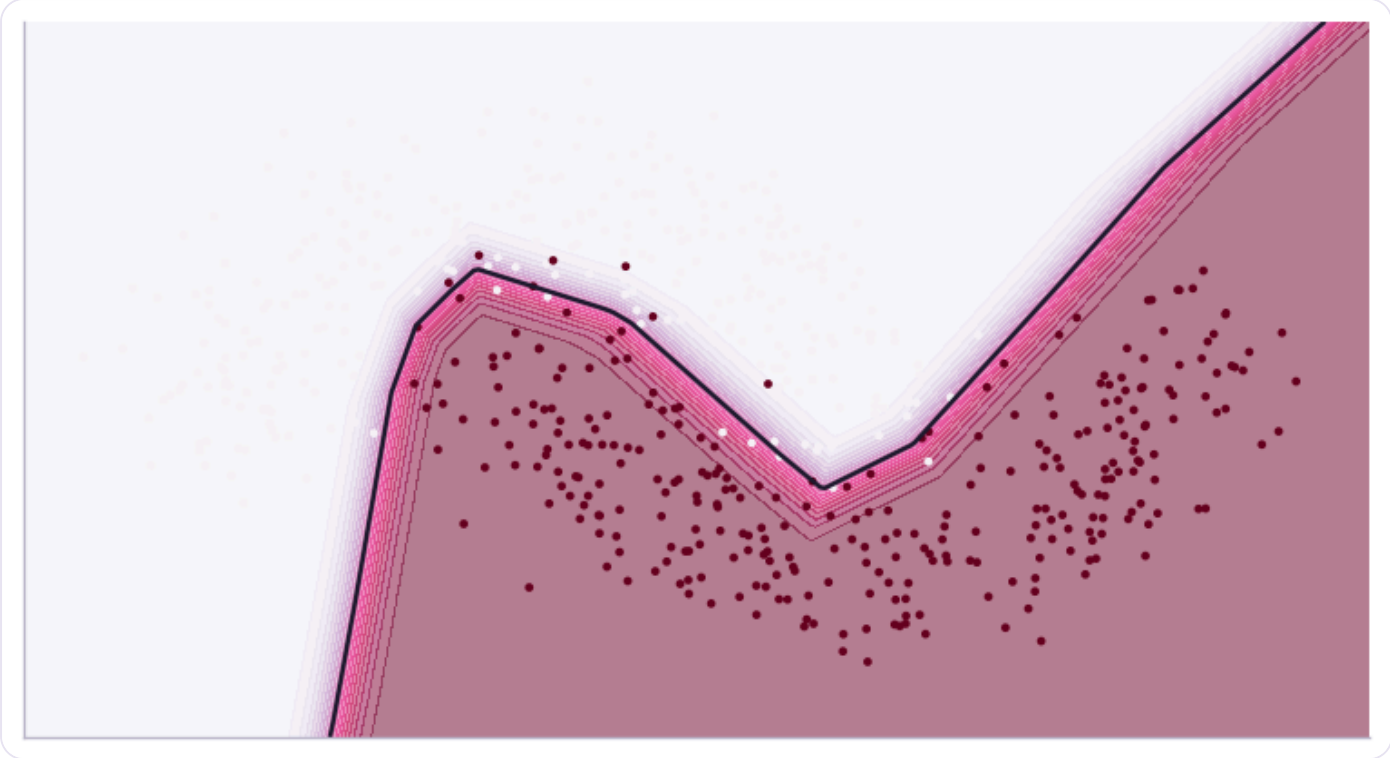
A network from scratch

Python

```
from sklearn.datasets import make_moons
X, y = make_moons(600, noise=0.2, random_state=0); y = y.reshape(-1, 1)
rng = np.random.default_rng(0)
W1, b1 = rng.normal(0, 1, (2, 16)) * np.sqrt(2 / 2), np.zeros((1, 16))          # He
initialisation
W2, b2 = rng.normal(0, 1, (16, 1)) * np.sqrt(2 / 16), np.zeros((1, 1))
sigmoid = lambda z: 1 / (1 + np.exp(-z))
lr, losses = 0.5, []
for epoch in range(2000):
    z1 = X @ W1 + b1; a1 = np.maximum(0, z1)                    # forward: hidden layer
(ReLU)
    p = sigmoid(a1 @ W2 + b2)                                    # forward: output
(sigmoid)
    loss = -np.mean(y * np.log(p + 1e-9) + (1 - y) * np.log(1 - p + 1e-9));
losses.append(loss)
    dz2 = (p - y) / len(X)                                       # backward: output
gradient
    dW2, db2 = a1.T @ dz2, dz2.sum(0, keepdims=True)
    dz1 = (dz2 @ W2.T) * (z1 > 0)                                # chain rule through ReLU
    dW1, db1 = X.T @ dz1, dz1.sum(0, keepdims=True)
    W1 -= lr * dW1; b1 -= lr * db1; W2 -= lr * dW2; b2 -= lr * db2
acc = ((p > 0.5) == y).mean()
print(f"loss: epoch 0 {losses[0]:.3f} → epoch 200 {losses[200]:.3f} → epoch 2000
{losses[-1]:.3f} | training accuracy {acc:.3f}")
print("parameters:", W1.size + b1.size + W2.size + b2.size)
```

OUTPUT

```
loss: epoch 0 0.719 → epoch 200 0.196 → epoch 2000 0.063 | training accuracy 0.977
parameters: 65
```



Decision boundary learned by the 16-neuron network written above: a curved boundary that a linear model cannot draw.

These 30 lines are the core of every deep-learning library; frameworks add automatic differentiation, GPUs, better optimisers and building blocks.

MLPs in scikit-learn

Python

```
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
digits = load_digits()
Xa, Xb, ya, yb = train_test_split(digits.data, digits.target, test_size=0.3,
stratify=digits.target, random_state=0)
for name, m in [("logistic regression", make_pipeline(StandardScaler(),
LogisticRegression(max_iter=3000))),
                ("MLP (64)", make_pipeline(StandardScaler(), MLPClassifier((64,),
max_iter=500, random_state=0))),
                ("MLP (128, 64), early stopping", make_pipeline(StandardScaler(),
MLPClassifier((128, 64), alpha=1e-3, early_stopping=True, max_iter=500, random_state=0)))]:
    m.fit(Xa, ya); print(f"{name:32s} test accuracy {m.score(Xb, yb):.3f}")
```

OUTPUT

```
logistic regression              test accuracy 0.972
MLP (64)                         test accuracy 0.974
MLP (128, 64), early stopping    test accuracy 0.974
```

Python

```python
import pandas as pd
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.ensemble import HistGradientBoostingClassifier
tc = pd.read_csv("telecom_churn.csv"); Xc = tc.drop(columns=["customer_id", "churn",
"total_charges"]); yc = (tc.churn == "Yes").astype(int)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months",
"monthly_charges", "support_calls"]), ("c", OneHotEncoder(), ["contract",
"internet_service", "payment_method"])])
for name, est in [("logistic regression", LogisticRegression(max_iter=2000)), ("MLP (32,
16)", MLPClassifier((32, 16), alpha=1e-2, early_stopping=True, max_iter=1000,
random_state=0)),
                  ("gradient boosting", HistGradientBoostingClassifier(max_iter=200,
learning_rate=0.05, random_state=0))]:
    s = cross_val_score(make_pipeline(pre, est), Xc, yc, cv=5, scoring="roc_auc");
print(f"churn {name:20s} AUC {s.mean():.3f}")
```

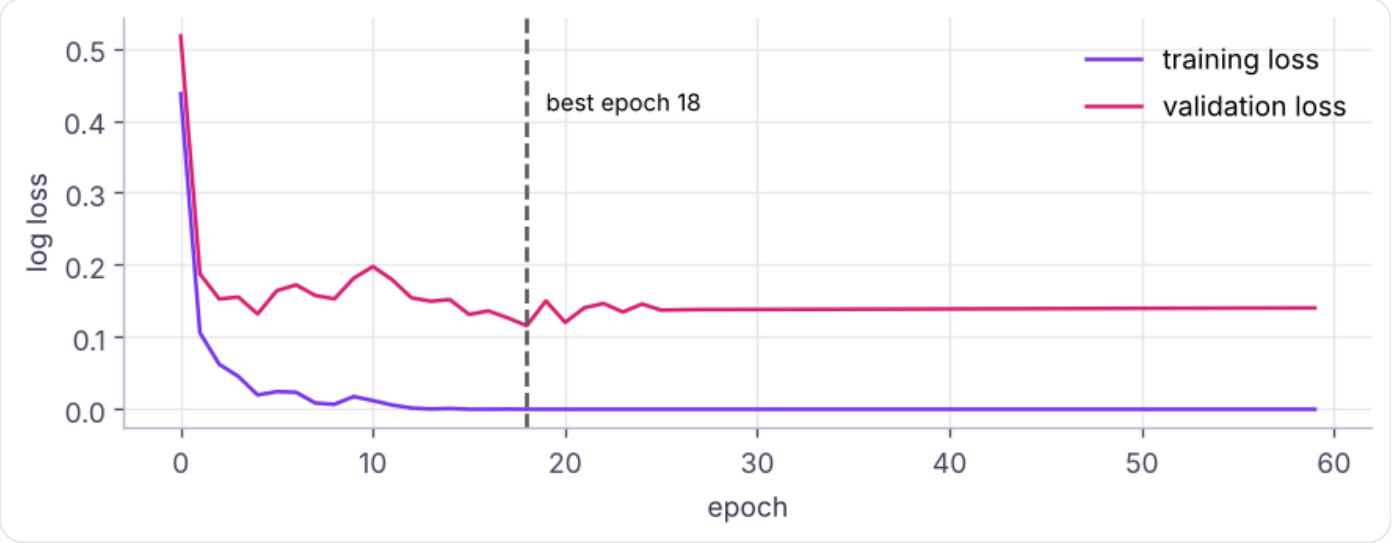
OUTPUT

```
churn logistic regression  AUC 0.815
churn MLP (32, 16)         AUC 0.802
churn gradient boosting    AUC 0.782
```

Neural networks win on **unstructured data** (images, text, audio). On small tabular data, logistic regression and gradient boosting usually match or beat them with less tuning.

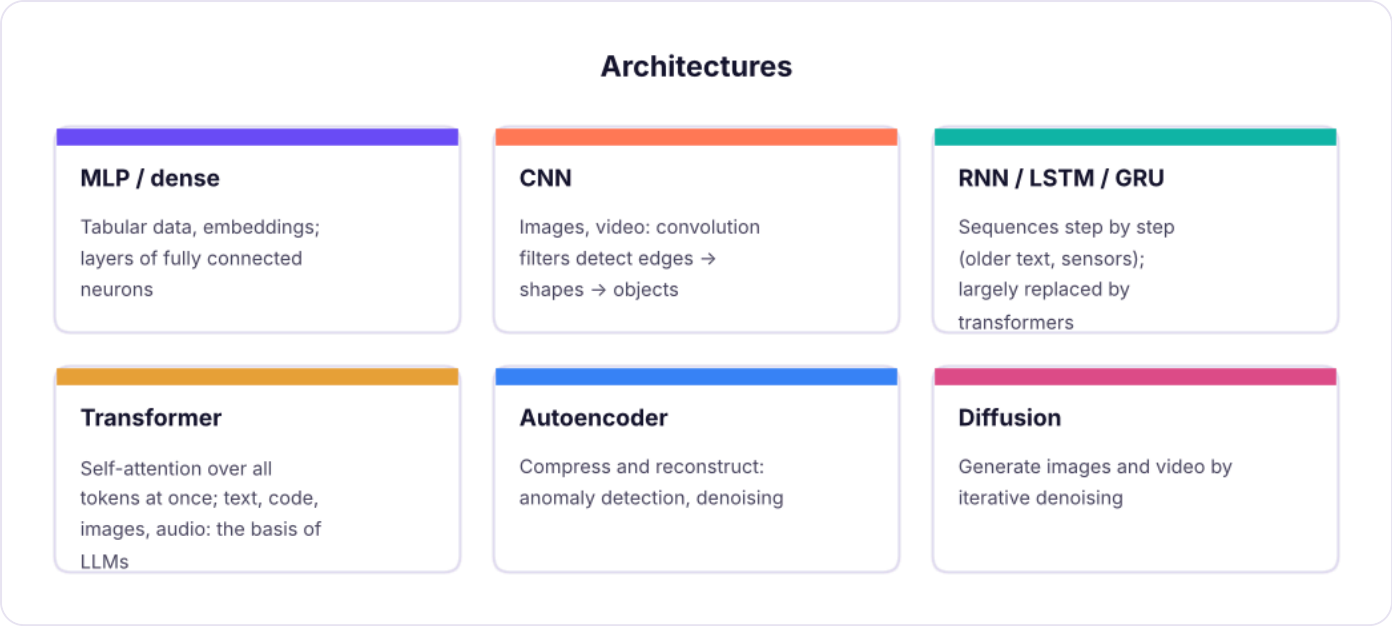
Training practice

Technique	Purpose
Scale inputs	Stable, fast optimisation
Learning rate (and schedules)	The single most important setting
Mini-batches (32–512)	Efficient GPU use, regularising noise
Epochs + **early stopping**	Stop when validation loss stops improving
Dropout	Randomly switch off neurons in training: reduces overfitting
Weight decay (L2)	Penalise large weights (`alpha` in scikit-learn)
Batch / layer normalisation	Stabilise activations, allow higher learning rates
Data augmentation	Flip/crop images, paraphrase text: more data for free
Transfer learning	Start from a pre-trained model and fine-tune: the default today



Training vs validation loss: when validation loss starts rising the network is overfitting; early stopping keeps the best epoch.

Deep-learning architectures



The main architecture families and their typical data.

Transformers (Vaswani et al., 2017, "Attention Is All You Need") let every token look at every other token through **self-attention**, train in parallel on GPUs and scale to billions of parameters. Pre-trained on huge text corpora to predict the next token, then instruction-tuned and aligned with human feedback, they became **large language models** (GenAI track).

PyTorch and Keras

Python

```python
# PyTorch (pip install torch): the research and industry standard
import torch, torch.nn as nn
model = nn.Sequential(nn.Linear(64, 128), nn.ReLU(), nn.Dropout(0.2), nn.Linear(128, 10))
opt = torch.optim.AdamW(model.parameters(), lr=1e-3, weight_decay=1e-4)
loss_fn = nn.CrossEntropyLoss()
for epoch in range(20):
    for xb, yb in train_loader:                # mini-batches
        opt.zero_grad()
        loss = loss_fn(model(xb), yb)
        loss.backward()                         # backpropagation (autograd)
        opt.step()                              # Adam update
```

Python

```python
# Keras 3 (pip install keras): high-level API on TensorFlow, JAX or PyTorch
import keras
model = keras.Sequential([keras.layers.Dense(128, activation="relu"),
keras.layers.Dropout(0.2), keras.layers.Dense(10, activation="softmax")])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy", metrics=
["accuracy"])
model.fit(X_train, y_train, validation_split=0.2, epochs=50, callbacks=
[keras.callbacks.EarlyStopping(patience=5, restore_best_weights=True)])
```

Python

```python
# Transfer learning with Hugging Face (pip install transformers): fine-tune a pre-trained
model for sentiment
from transformers import pipeline
clf = pipeline("sentiment-analysis", model="distilbert-base-uncased-finetuned-sst-2-
english")
clf(["Battery barely lasts half a day", "Display is bright and sharp"])
```

Free GPUs for practice: Google Colab and Kaggle notebooks.

Summary

KEY TAKEAWAYS

- A neuron = weighted sum + activation; layers of neurons with non-linear activations learn complex functions.
- Training = forward pass, loss, backpropagation (chain rule), gradient-descent updates (Adam) over epochs and mini-batches.
- Use ReLU/GELU in hidden layers, sigmoid/softmax outputs for classification; scale inputs; regularise with early stopping, dropout and weight decay.
- Neural networks excel on images, text and audio; on small tabular data, gradient boosting and logistic regression are strong competitors.
- CNNs for images, transformers for sequences and language (LLMs); PyTorch, Keras and Hugging Face are the main tools; transfer learning is the default.

Quick quiz

6 questions · tap an answer to check it

1. What happens without non-linear activation functions?

- **A** The whole network collapses to a linear model
- **B** It trains faster with better accuracy
- **C** It becomes a decision tree
- **D** Nothing changes

Answer: A. Non-linearity is what gives depth its power.

2. What does backpropagation compute?

- **A** Gradients of the loss with respect to every weight, using the chain rule
- **B** The predictions
- **C** The learning rate
- **D** The number of layers

Answer: A. Gradient descent then updates the weights.

3. Which output activation for a 10-class problem?

- **A** ReLU
- **B** Softmax
- **C** None
- **D** tanh

Answer: B. Probabilities that sum to 1.

4. **Validation loss starts rising while training loss keeps falling. What is happening?**

- **A** Overfitting; stop early or add regularisation
- **B** Underfitting
- **C** The learning rate is perfect
- **D** Data leakage for sure

Answer: A. Early stopping keeps the best epoch.

5. **Which architecture is the basis of LLMs?**

- **A** CNN
- **B** RNN
- **C** Transformer
- **D** K-means

Answer: C. Self-attention over tokens.

6. **For a 2,000-row tabular churn dataset, which is usually a strong first choice?**

- **A** Logistic regression or gradient boosting
- **B** A 50-layer CNN
- **C** An LLM fine-tune
- **D** An autoencoder

Answer: A. Deep learning shines on unstructured data.

EXERCISE

Practice

(1) Change the hidden layer size of the NumPy network to 4 and 64: how do loss and boundary change? (2) Train an `MLPRegressor` on house prices and compare MAE with gradient boosting. (3) In Colab, train the PyTorch model above on the digits data.

Show solution (worked answer)

Python

```
from sklearn.neural_network import MLPRegressor
from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"]); h =
h[h.price_lakh.between(5, 300)]
Xh = pd.get_dummies(h.drop(columns=["house_id", "price_lakh"]),
drop_first=True).astype(float)
a, b, c, d = train_test_split(Xh, h.price_lakh, test_size=0.2, random_state=0)
mlp_r = make_pipeline(StandardScaler(), MLPRegressor((64, 32), alpha=1e-3,
max_iter=3000, early_stopping=True, random_state=0)).fit(a, c)
print(mean_absolute_error(d, mlp_r.predict(b)), mean_absolute_error(d,
HistGradientBoostingRegressor().fit(a, c).predict(b)))
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain how a neural network learns. It makes predictions with its current weights, measures the error with a loss function, computes how each weight affected the error through backpropagation, and adjusts the weights slightly in the direction that reduces the error, repeating over many batches and epochs.

Q: What is the vanishing gradient problem? In deep networks with sigmoid/tanh, gradients shrink as they pass backward through layers, so early layers learn very slowly; ReLU, residual connections and normalisation fix it.

Q: What is transfer learning? Reusing a model pre-trained on a large dataset as the starting point for a new task, fine-tuning some or all layers with much less data.

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